Universidade Federal de Santa Catarina
Centro Tecnológico - CTC
Programa de Pós-Graduação em Engenharia Elétrica
Instituto de Eletrônica de Potência – INEP

EEL410275 – Análise Estática, Modelagem e Controle de Conversores Básicos

AULA 5 – MODELAGEM ORIENTADA AO CONTROLE: POR SUBSTITUIÇÃO DE FONTE

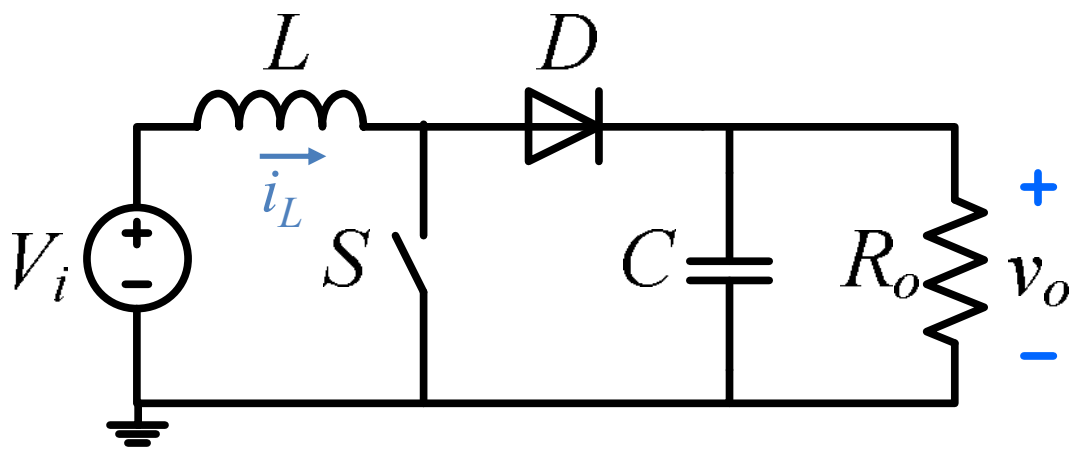
Profª Jéssika Melo de Andrade, Dra. Eng.
melo.jessika@ufsc.br

Florianópolis, 2024

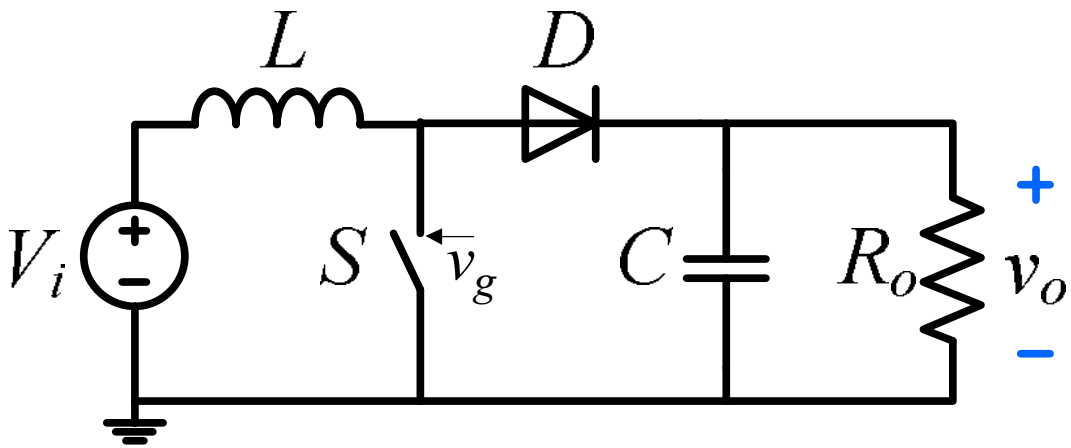
OBJETIVOS

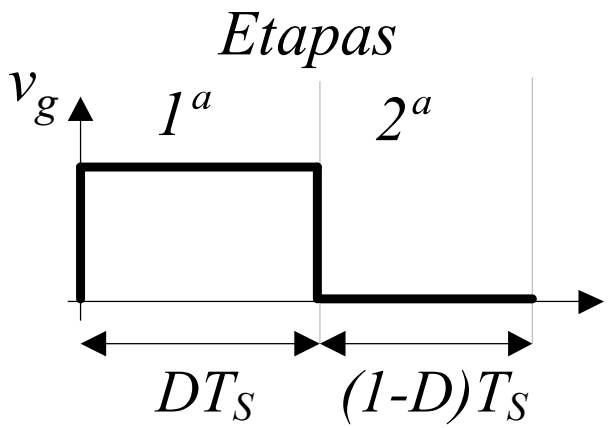
1) Determinar as funções de transferência:

$$\begin{cases} G_{vd}(s) = \dfrac{\hat{v}_o}{\hat{d}} \\ G_{id}(s) = \dfrac{\hat{i}_L}{\hat{d}} \\ G_{vi}(s) = \dfrac{\hat{v}_o}{\hat{i}_L} \end{cases}$$



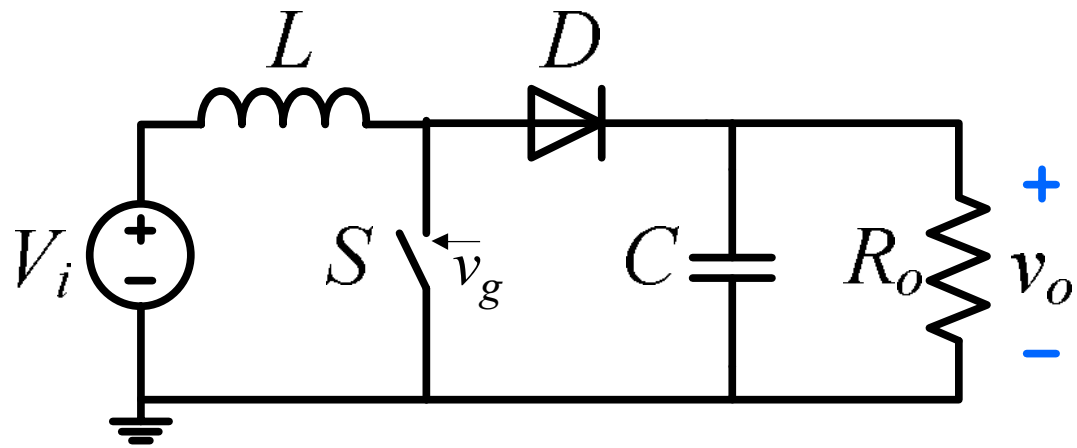
MODELO COMUTADO

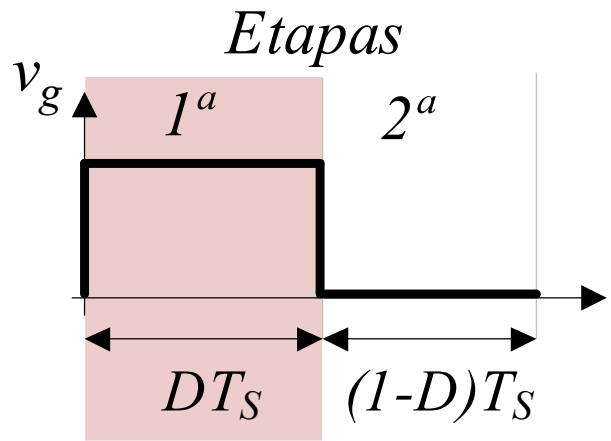




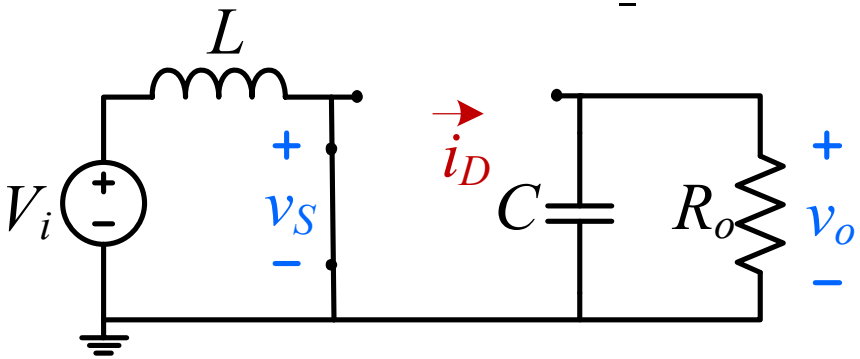
- **SUBSTITUIR:**
  - Interruptor por uma fonte de tensão controlada;
  - Diodo por uma fonte de corrente controlada.

MODELO COMUTADO



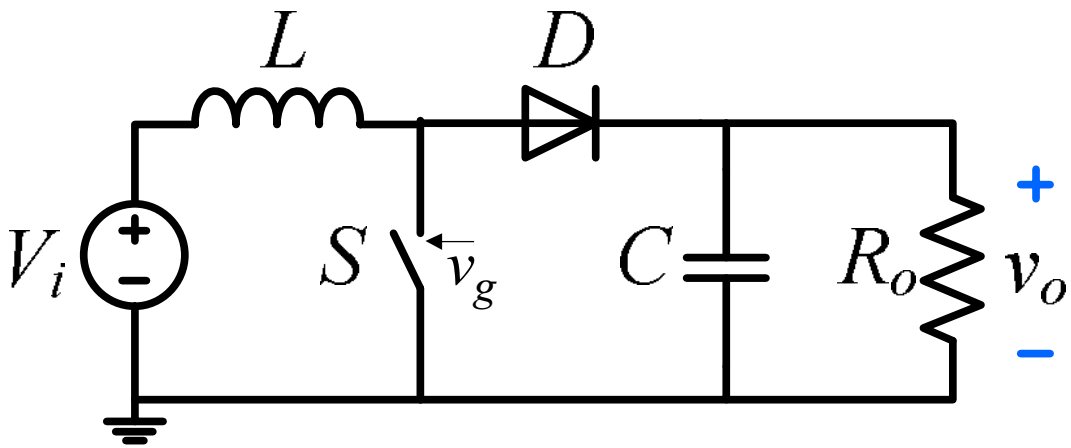


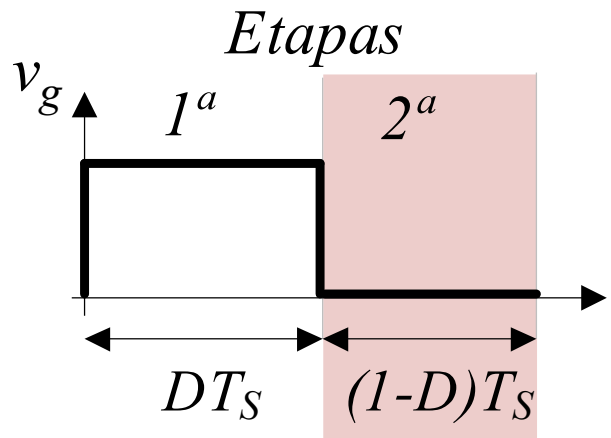
1ª Etapa: DT_s



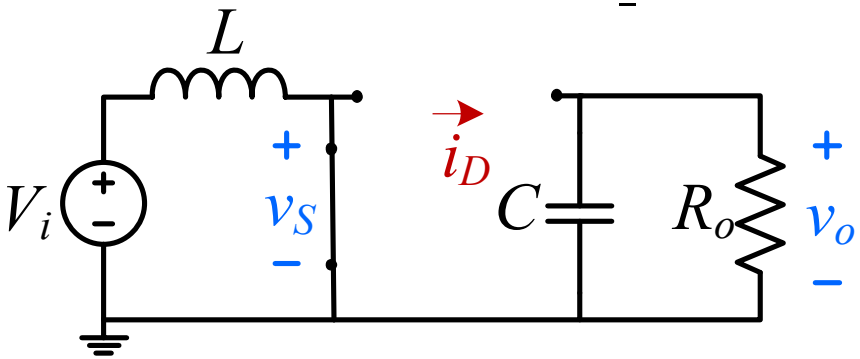
$$\begin{cases} v_S = 0 \\ i_D = 0 \end{cases}$$

MODELO COMUTADO



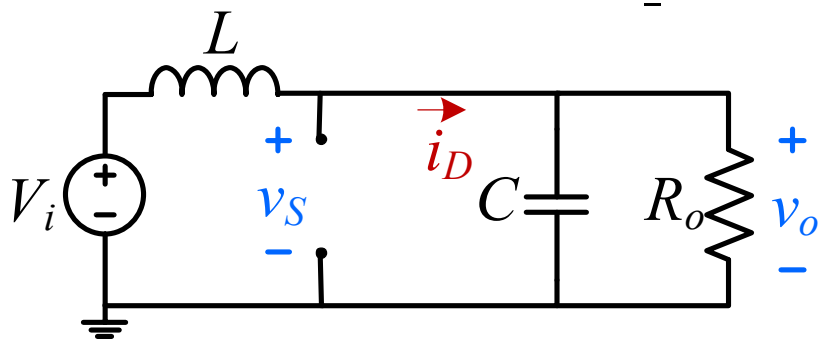


1ª Etapa: DT_s



$$\begin{cases} v_S = 0 \\ i_D = 0 \end{cases}$$

2ª Etapa: $(1\text{-}D)T_s$



$$\begin{cases} v_S = v_o \\ i_D = i_L \end{cases}$$

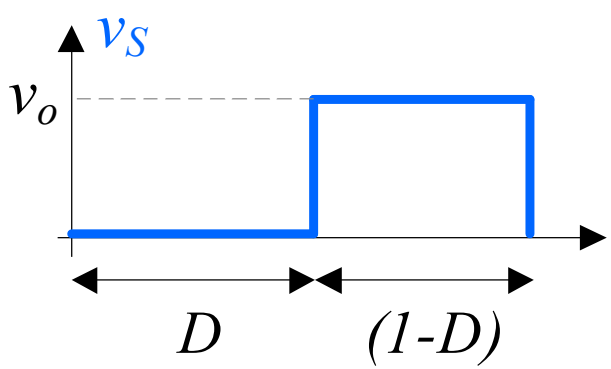
MODELO COMUTADO

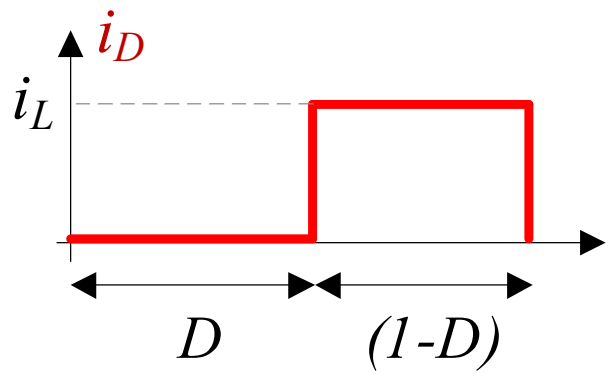
1ª Etapa: DT_s

$$\begin{cases} v_S = 0 \\ i_D = 0 \end{cases}$$

2ª Etapa: $(1\text{-}D)T_s$

$$\begin{cases} v_S = v_o \\ i_D = i_L \end{cases}$$

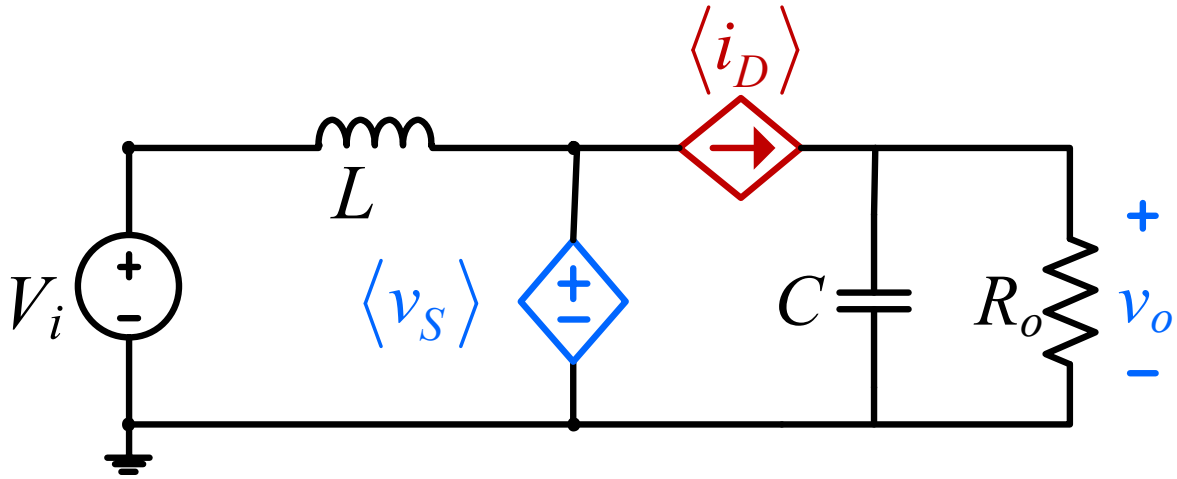




$$\langle v_S \rangle = v_o (1 - D)$$

$$\langle i_D \rangle = i_L (1 - D)$$

MODELO MÉDIO DE GRANDES SINAIS



$$\langle v_S \rangle = v_o (1 - D)$$
$$\langle i_D \rangle = i_L (1 - D) \qquad (1)$$

Ponto de operação + pequena perturbação:

$$\begin{cases} D = D + \hat{d} \\ i_L = I_L + \hat{i}_L \\ v_o = V_o + \hat{v}_o \\ i_D = I_D + \hat{i}_D \\ v_S = V_S + \hat{v}_S \end{cases} \qquad (2)$$

MODELO MÉDIO DE GRANDES SINAIS

$$\boxed{\begin{aligned}\langle v_S \rangle &= v_o\left(1-D\right)\\ \langle i_D \rangle &= i_L\left(1-D\right)\end{aligned}} \quad (1)$$

$$\boxed{\begin{cases} D = D + \hat{d} \\ i_L = I_L + \hat{i}_L \\ v_o = V_o + \hat{v}_o \\ i_D = I_D + \hat{i}_D \\ v_S = V_S + \hat{v}_S \end{cases}} \quad (2)$$

$$(2) \rightarrow (1)$$

$$V_S + \hat{v}_S = \left(V_o + \hat{v}_o\right)\left(1 - D - \hat{d}\right) \qquad \begin{cases} V_S = V_o\left(1-D\right) \\ \hat{v}_S = \hat{v}_o\left(1-D\right) - V_o\hat{d} \end{cases}$$

$$I_D + \hat{i}_D = \left(I_L + \hat{i}_L\right)\left(1 - D - \hat{d}\right) \qquad \begin{cases} I_D = I_L\left(1-D\right) \\ \hat{i}_D = \hat{i}_L\left(1-D\right) - I_L\hat{d} \end{cases}$$

MODELO MÉDIO DE GRANDES SINAIS

$$\begin{cases} V_S = V_o\left(1-D\right) \\ \hat{v}_S = \hat{v}_o\left(1-D\right) - V_o\hat{d} \end{cases}$$

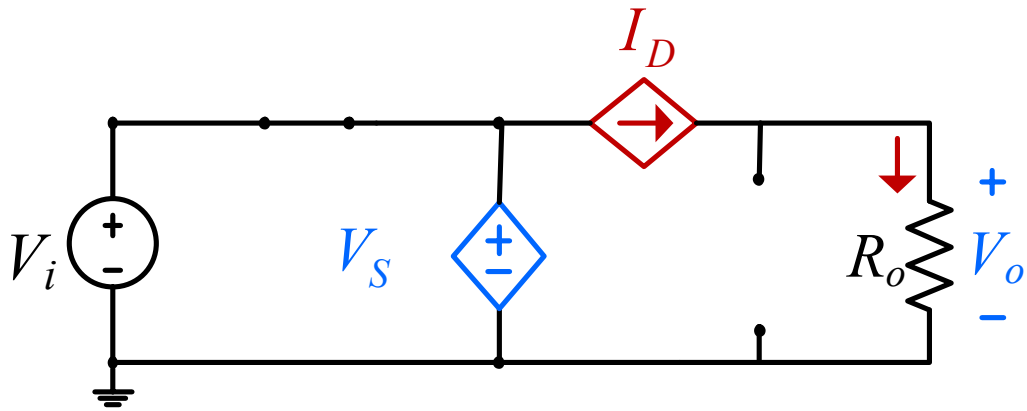
$$\begin{cases} I_D = I_L\left(1-D\right) \\ \hat{i}_D = \hat{i}_L\left(1-D\right) - I_L\hat{d} \end{cases}$$

Modelo CC

Modelo CA ou
Modelo de Pequenos Sinais

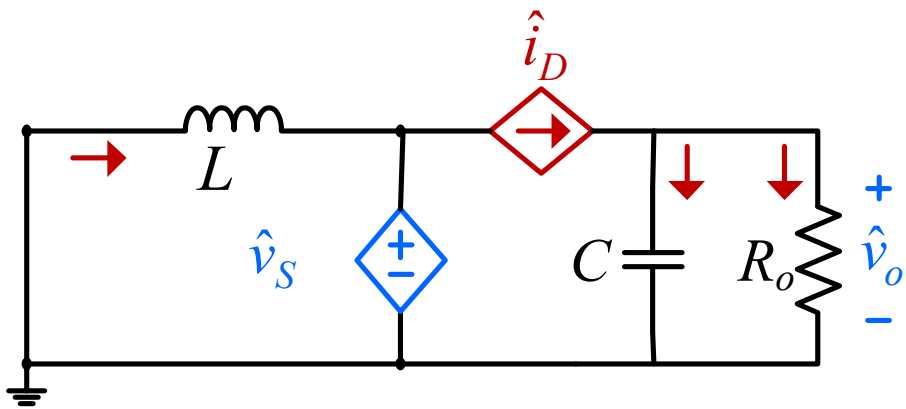
MODELOS

Modelo CC: descreve o conversor no ponto de operação.



$$\begin{cases} V_S = V_o\left(1-D\right) \\ I_D = I_L\left(1-D\right) \end{cases}$$

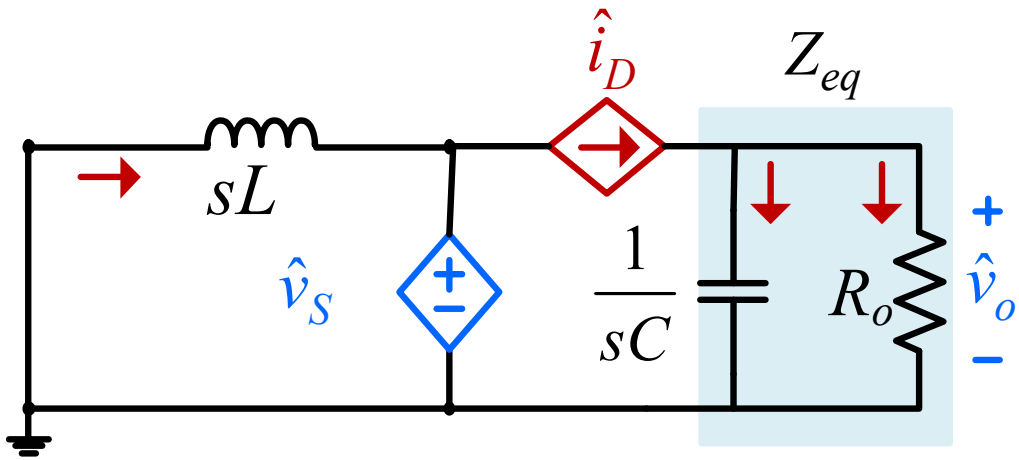
Modelo de pequenos sinais: descreve a dinâmica do conversor.



$$\begin{cases} \hat{v}_S = \hat{v}_o\left(1-D\right) - V_o\hat{d} \\ \hat{i}_D = \hat{i}_L\left(1-D\right) - I_L\hat{d} \end{cases}$$

ANÁLISE PARA FUNÇÃO DE TRANSFERÊNCIA G(s) = v_o/d

MODELO DE PEQUENOS SINAIS



$$\begin{cases} \hat{v}_S = \hat{v}_o(1-D) - V_o\hat{d} \\ \hat{i}_D = \hat{i}_L(1-D) - I_L\hat{d} \end{cases}$$

$$\boxed{G_{vd}(s) = \frac{\hat{v}_o}{\hat{d}}}$$

$$(1) \quad Z_{eq} = \frac{R_o}{R_oCs+1}$$

$$(2) \quad \hat{v}_o = Z_{eq}\hat{i}_D = Z_{eq}\left[\hat{i}_L(1-D) - I_L\hat{d}\right]$$

$$\hat{v}_L + \hat{v}_S = 0$$

$$\hat{v}_L + \hat{v}_o(1-D) - V_o\hat{d} = 0$$

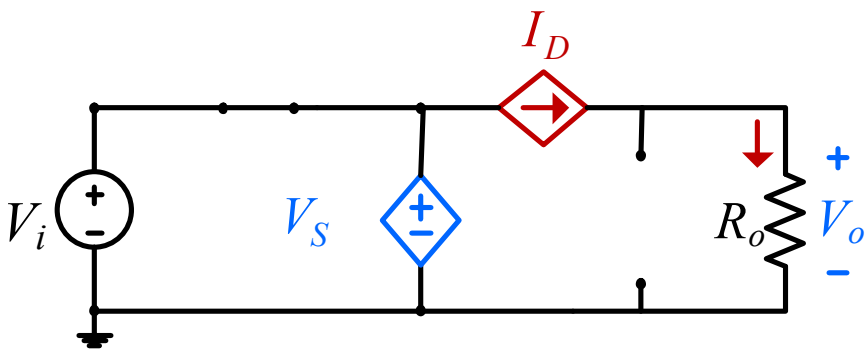
$$sL\hat{i}_L + \hat{v}_o(1-D) - V_o\hat{d} = 0$$

$$(3) \quad \hat{i}_L = \frac{V_o\hat{d} - \hat{v}_o(1-D)}{sL}$$

(3) em (2)

$$\boxed{\frac{\hat{v}_o}{\hat{d}} = \frac{R_oV_o(1-D) - I_LR_oLs}{R_oCLs^2 + sL + R_o(1-D)^2}}$$

MODELO CC



$$\begin{cases} V_S = V_o\left(1-D\right) \\ I_D = I_L\left(1-D\right) \end{cases}$$

$$I_D = I_o$$

$$I_L\left(1-D\right) = \frac{V_o}{R_o}$$

$$I_L\left(1-D\right) = \frac{V_i}{\left(1-D\right)R_o}$$

$$I_L = \frac{V_i}{\left(1-D\right)^2 R_o}$$

$$\frac{V_o}{V_i} = \frac{1}{1-D}$$

$$V_o = \frac{V_i}{\left(1-D\right)}$$

FUNÇÃO DE TRANSFERÊNCIA

$$G_{vd}(s) = \frac{\hat{v}_o}{\hat{d}}$$

$$I_L = \frac{V_i}{(1-D)^2 R_o}$$

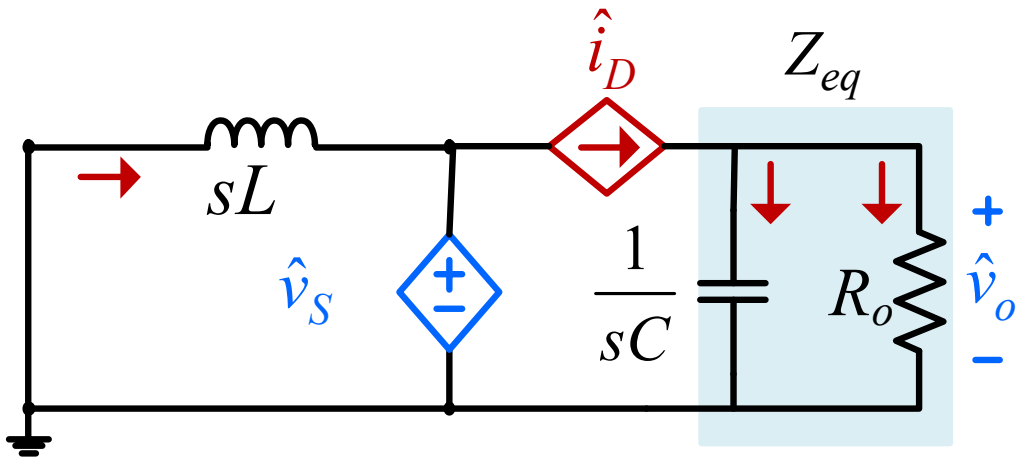
$$V_o = \frac{V_i}{(1-D)}$$

$$\frac{\hat{v}_o}{\hat{d}} = \frac{R_o V_o (1-D) - I_L R_o L s}{R_o C L s^2 + sL + R_o (1-D)^2}$$

$$\frac{\hat{v}_o}{\hat{d}} = \frac{\dfrac{V_i}{LC} - \dfrac{V_i s}{(1-D)^2 R_o C}}{s^2 + \dfrac{s}{R_o C} + \dfrac{(1-D)^2}{LC}}$$

ANÁLISE PARA FUNÇÃO DE TRANSFERÊNCIA G(s) = i_L/d

MODELO DE PEQUENOS SINAIS



$$\begin{cases} \hat{v}_S = \hat{v}_o(1-D) - V_o\hat{d} \\ \hat{i}_D = \hat{i}_L(1-D) - I_L\hat{d} \end{cases}$$

$$(1) \quad Z_{eq} = \frac{R_o}{R_oCs+1}$$

$$(2) \quad \hat{v}_o = Z_{eq}\hat{i}_D = Z_{eq}\left[\hat{i}_L(1-D) - I_L\hat{d}\right]$$

$$\boxed{G_{id}(s) = \frac{\hat{i}_L}{\hat{d}}}$$

$$\hat{v}_L + \hat{v}_S = 0$$

$$\hat{v}_L + \hat{v}_o(1-D) - V_o\hat{d} = 0$$

$$(3) \quad sL\hat{i}_L + \hat{v}_o(1-D) - V_o\hat{d} = 0$$

(3) em (2)

$$\boxed{\frac{\hat{i}_L}{\hat{d}} = \frac{R_oI_L(1-D) + V_o + V_oR_oCs}{R_oCLs^2 + sL + R_o(1-D)^2}}$$

FUNÇÃO DE TRANSFERÊNCIA

$$G_{id}(s) = \frac{\hat{i}_L}{\hat{d}}$$

$$I_L = \frac{V_i}{(1-D)^2 R_o}$$

$$V_o = \frac{V_i}{(1-D)}$$

$$\frac{\hat{i}_L}{\hat{d}} = \frac{R_o I_L (1-D) + V_o + V_o R_o C s}{R_o C L s^2 + sL + R_o (1-D)^2}$$

$$\frac{\hat{i}_L}{\hat{d}} = \frac{\dfrac{2V_i}{(1-D) R_o C L} + \dfrac{V_i s}{(1-D) L}}{s^2 + \dfrac{s}{R_o C} + \dfrac{(1-D)^2}{LC}}$$

ANÁLISE PARA FUNÇÃO DE TRANSFERÊNCIA G(s) = v_o/i_L

FUNÇÃO DE TRANSFERÊNCIA

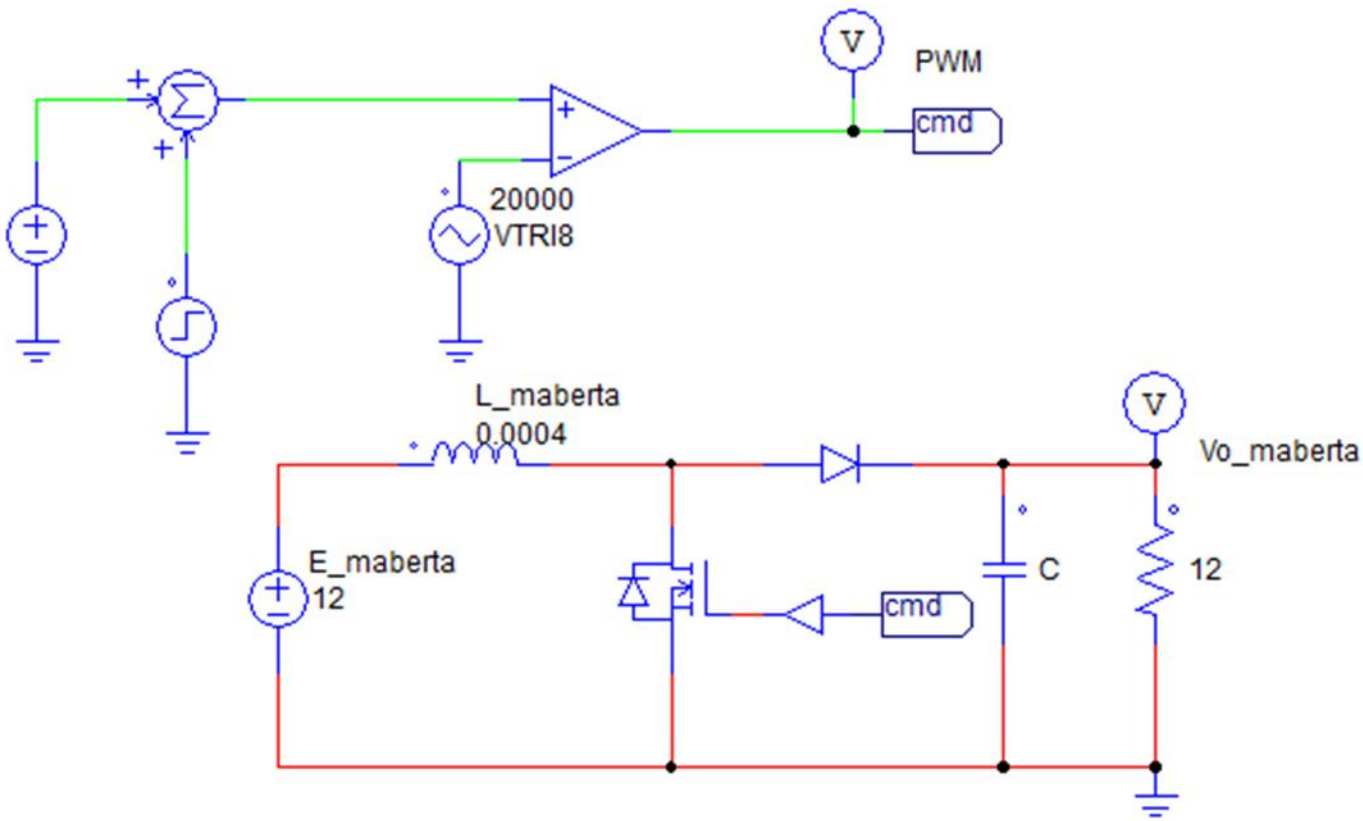
$$G_{vi}(s) = \frac{\hat{v}_o}{\hat{i}_L} = \frac{G_{vd}(\mathrm{s})}{G_{id}(\mathrm{s})}$$

$$G_{vi}(s) = \frac{\dfrac{\hat{v}_o}{\cancel{\hat{d}}}}{\dfrac{\hat{i}_L}{\cancel{\hat{d}}}} = \frac{\hat{v}_o}{\hat{i}_L}$$

VALIDAÇÃO DOS MODELOS

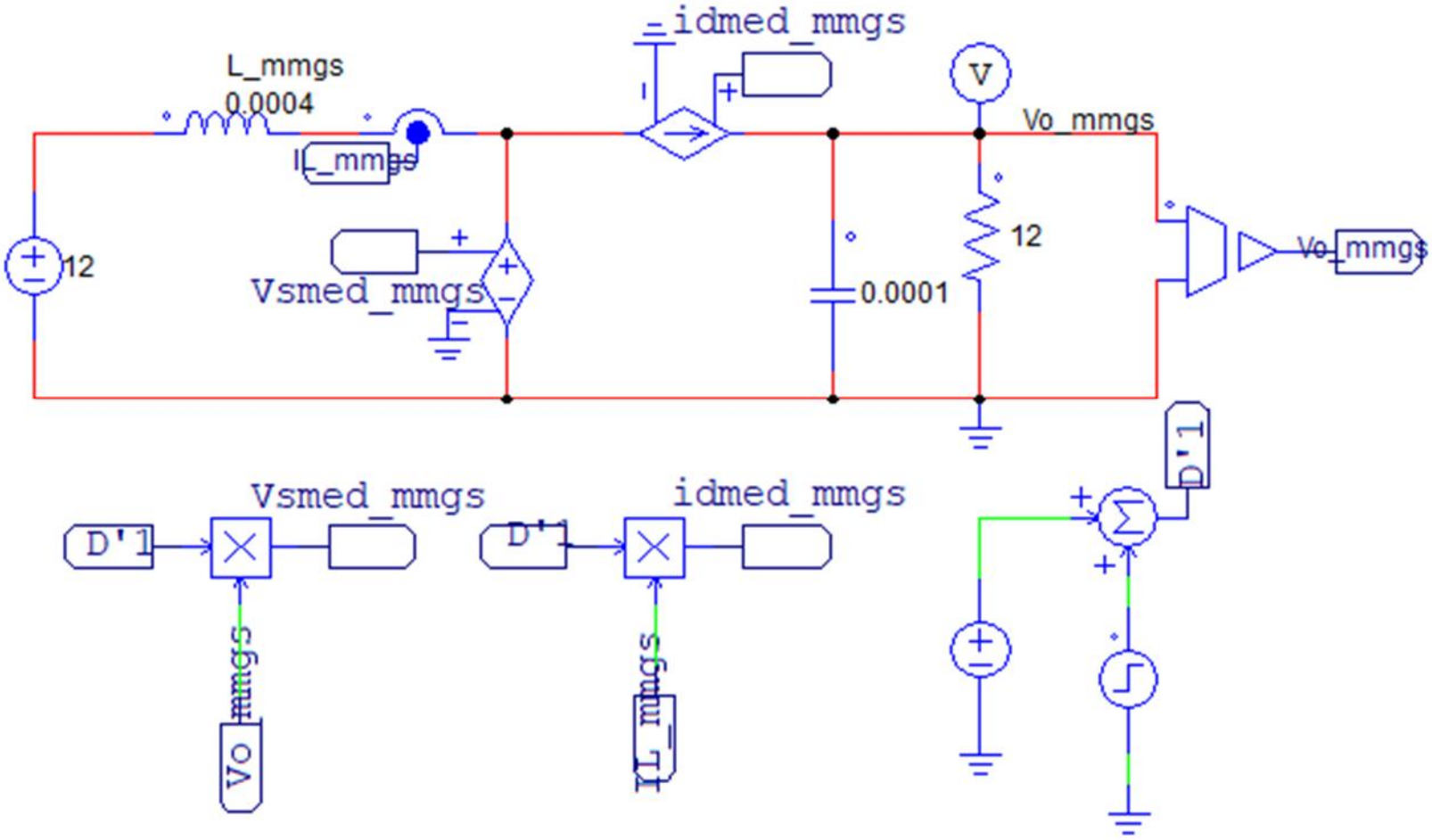
VALIDAÇÃO DOS MODELOS

Malha Aberta - Modelo Comutado



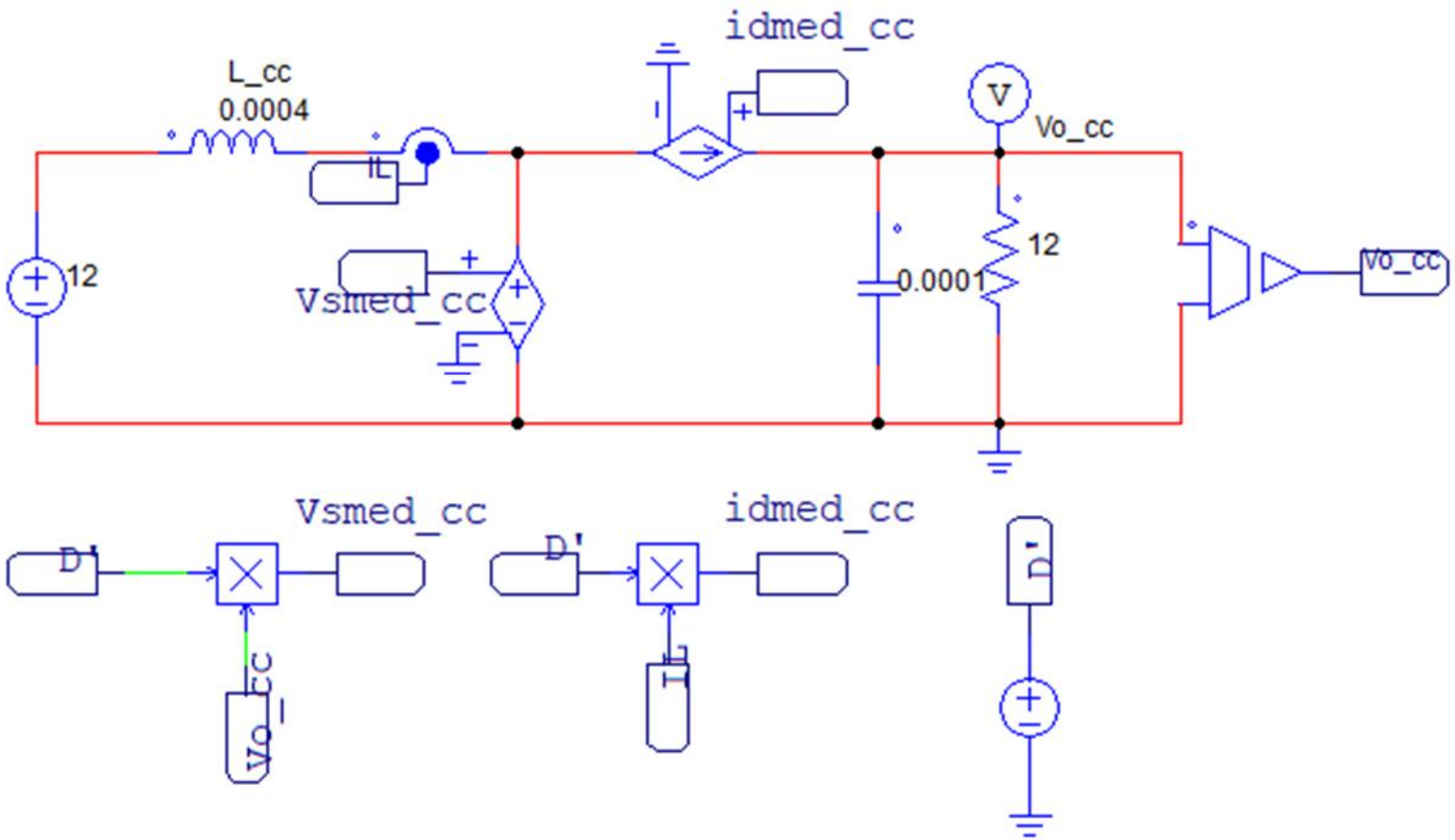
VALIDAÇÃO DOS MODELOS

Modelo Médio de Grandes Sinais



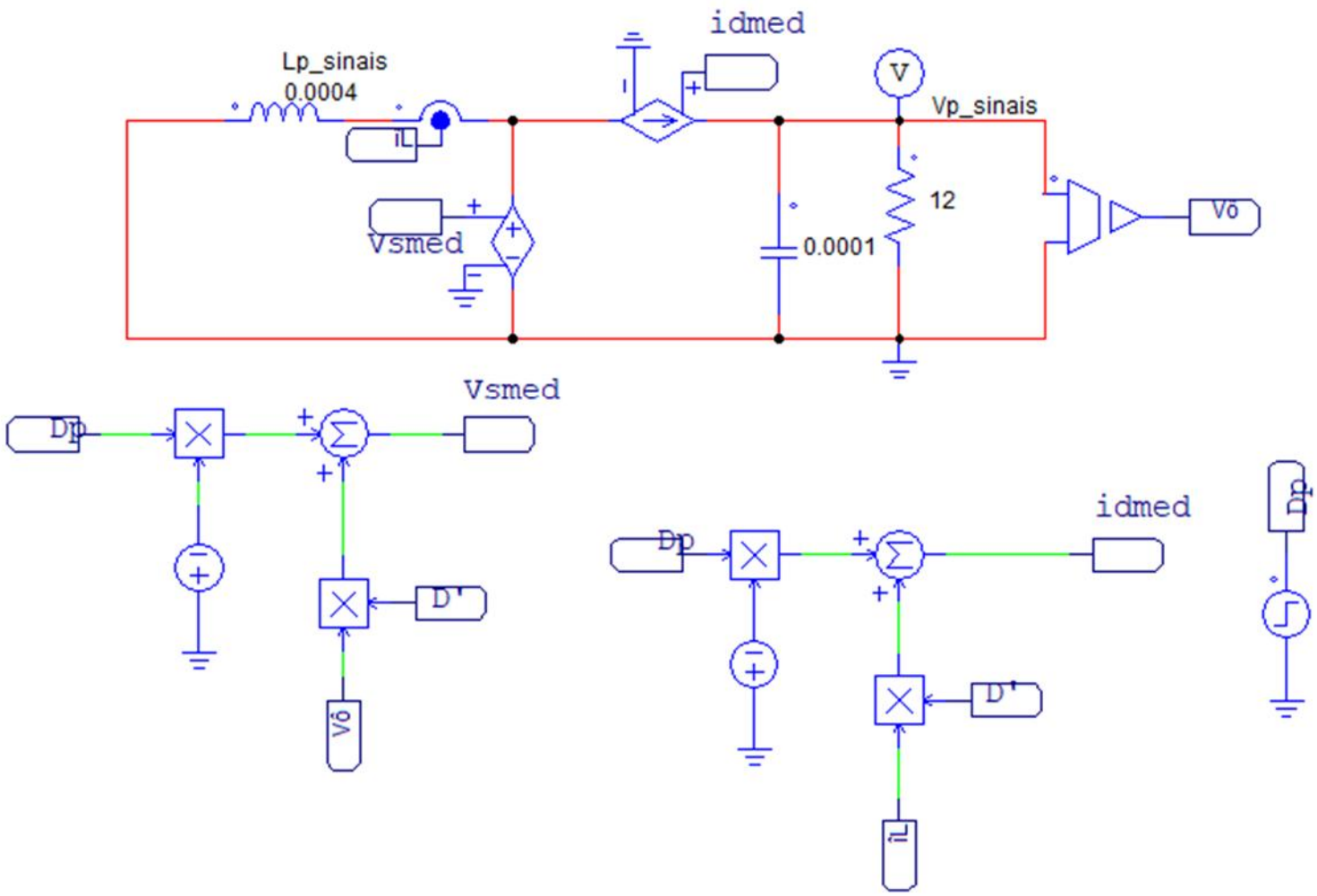
VALIDAÇÃO DOS MODELOS

Modelo CC

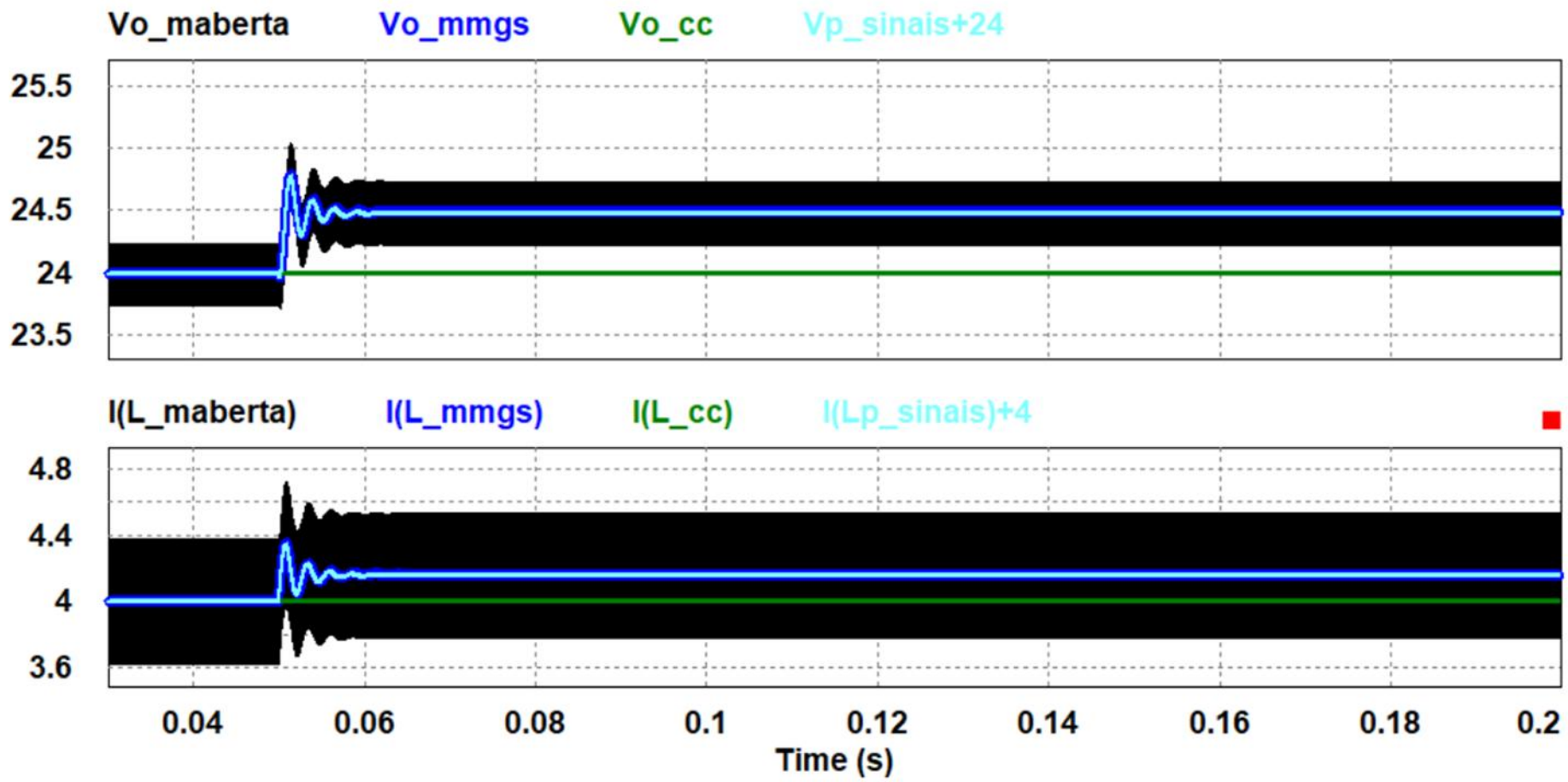


VALIDAÇÃO DOS MODELOS

Modelo de Pequenos Sinais



VALIDAÇÃO DOS MODELOS



VALIDAÇÃO DAS FUNÇÕES DE TRANSFERÊNCIA (FTs)

VALIDAÇÃO DAS FTs

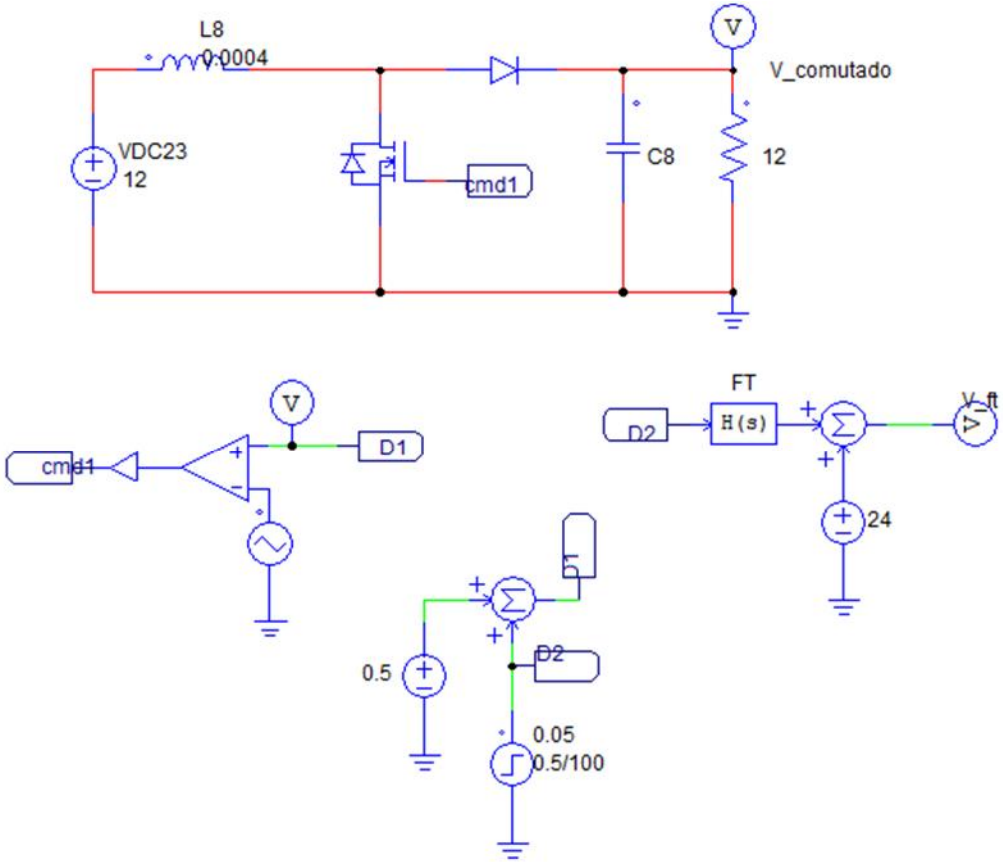
- As funções de transferência podem ser validadas no domínio do tempo ou da frequência.

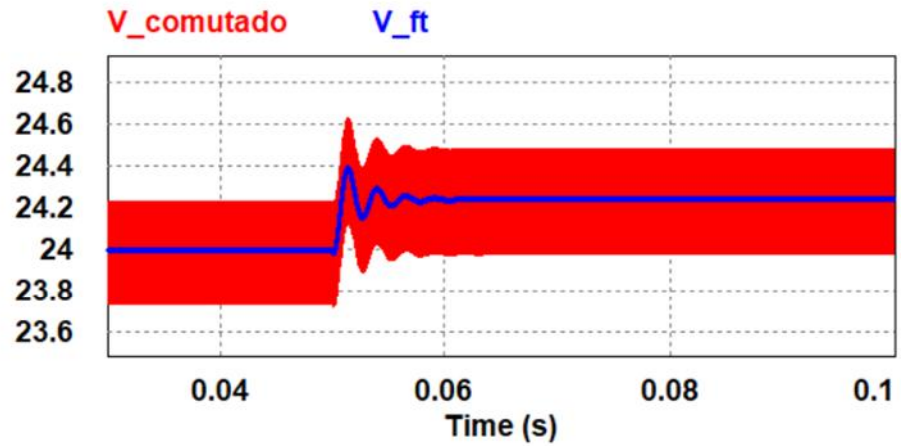
- No domínio do tempo é aplicado um degrau de razão cíclica no conversor;

- No domínio da frequência é aplicado no conversor variações de frequência e é avaliado o módulo e a fase da variável a ser controlada. Para isso se utiliza o bloco '*acsweep*' do PSIM.

VALIDAÇÃO DAS FTs

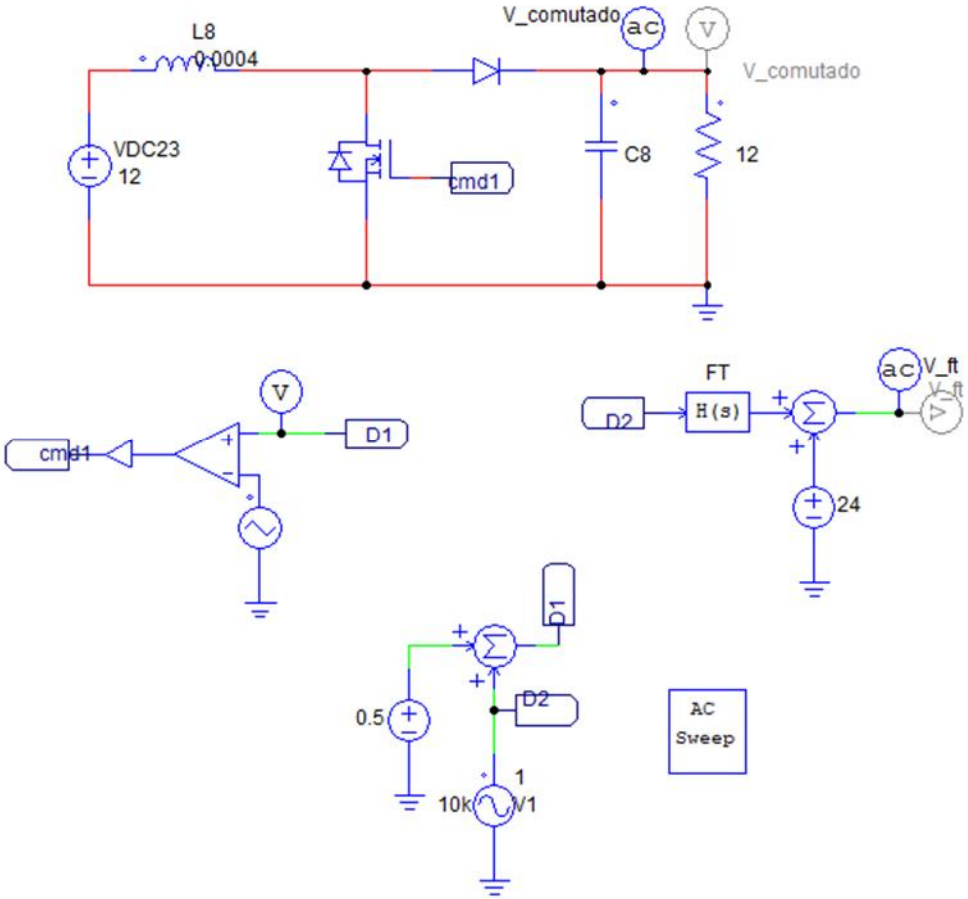
- **Exemplo: Domínio do Tempo – Validação de v_o/d**

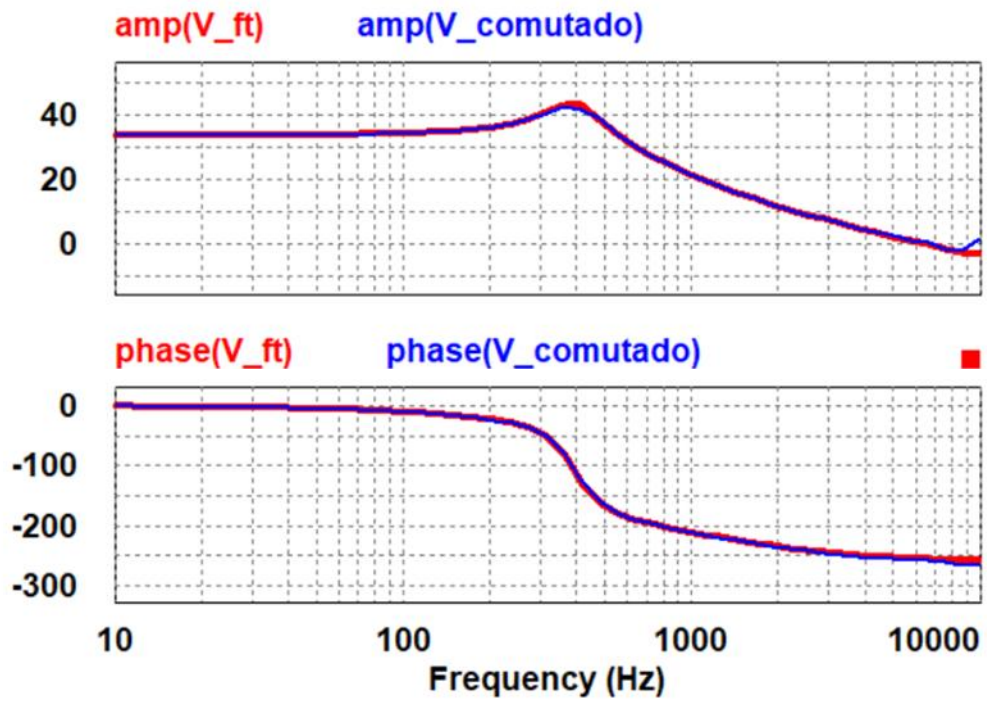




VALIDAÇÃO DAS FTs

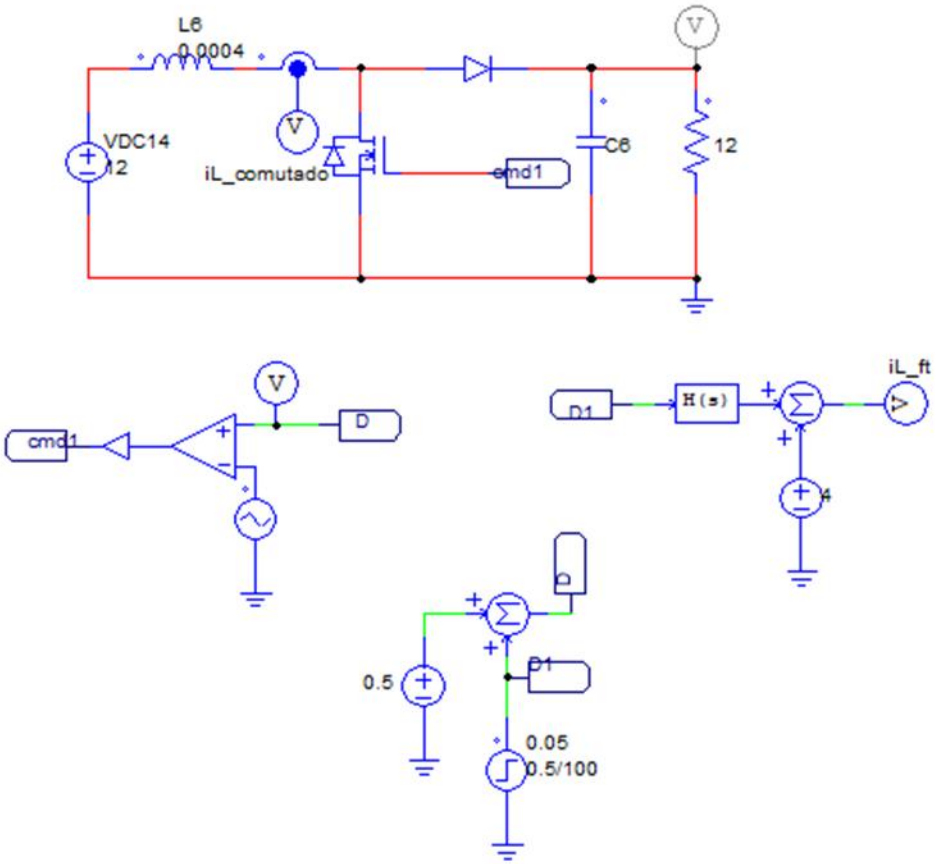
- **Exemplo: Domínio da Frequência – Validação de v_o/d**

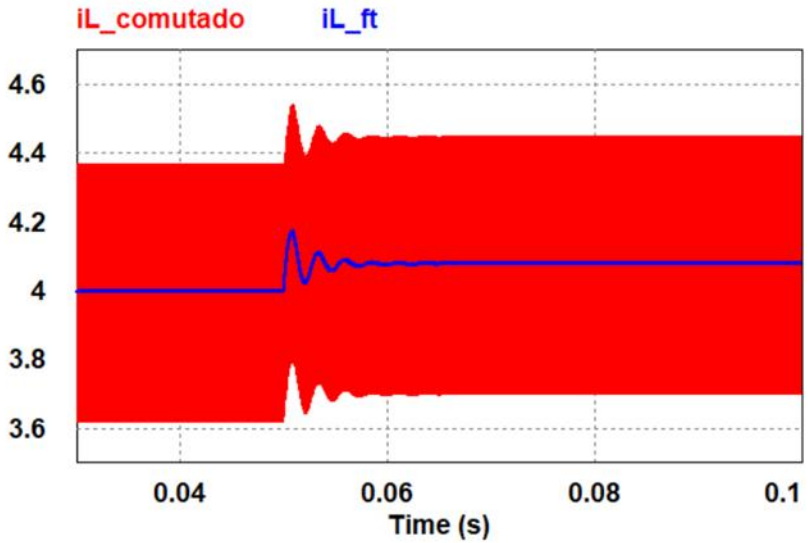




VALIDAÇÃO DAS FTs

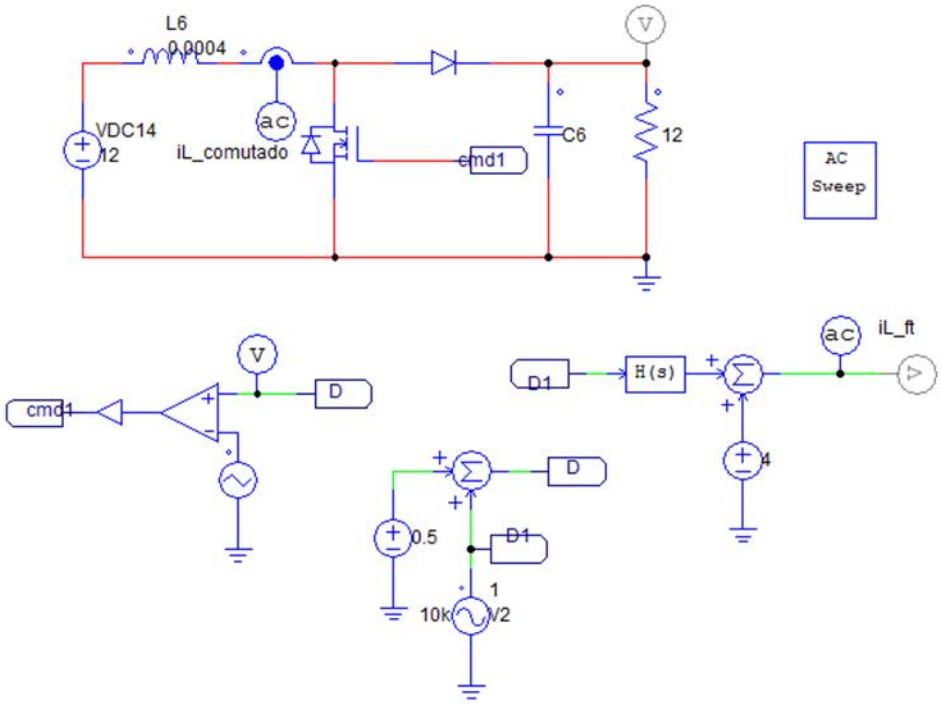
- **Exemplo: Domínio do Tempo – Validação de i_L/d**

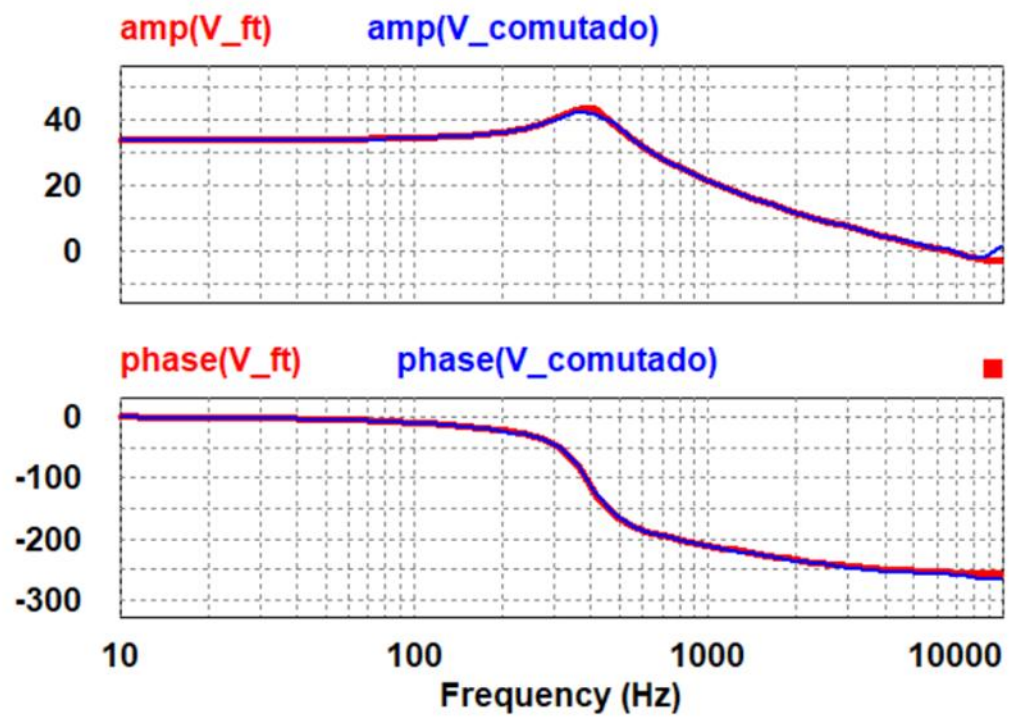




VALIDAÇÃO DAS FTs

- **Exemplo: Domínio da Frequência – Validação de i_L/d**





REFERÊNCIAS

[1] R. W. Erickson, D. Maksimovic, Fundamentals of power electronics, Massachusetts: Kluwer Academic Publishers, 2nd edition, New York, 2004.

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EEL410275 – Análise Estática, Modelagem e Controle de Conversores Básicos

ATÉ A PRÓXIMA AULA!

Profª Jéssika Melo de Andrade, Dra. Eng.

melo.jessika@ufsc.br

Florianópolis, 2024